

Veli — Frontend Design System

Cloud Parallax (Pvt) Ltd · v0.2 · 2026-05-01

1 Purpose

Canonical reference for visual and interaction patterns used in the Veli smartphone surface (Tamil-first PWA, Android wrap via Capacitor). Read before adding any UI. Pages reach for the primitives defined here rather than inline raw Tailwind classes for things the primitives already cover.

This is v0.2 — a civic-utilitarian system that replaces the prior shadcn-svelte zinc base. The shift was driven by a design exploration on claude.ai/design; the chosen aesthetic is “modern utilitarian / government-grade”, calibrated for rural Northern Province households on mid-range Android.

2 Four commitments

These are non-negotiable. Every design decision is judged against them.

Tamil	English	Detail
தமிழ் முதன்மை	Tamil-first	Designed in Tamil. English follows when it must, never as default. Display register: colloquial Jaffna Tamil, never literary, never Indian Tamil.
நம்பிக்கை	Trust by citation	Every claim links back to a government source or an audit trail. The platform never inflates a self-asserted artefact into something it is not.
குறை வளம்	Low-bandwidth	Works on a Rs. 18,000 Android with one bar of signal — or no signal at all. Service-worker offline cache is part of the contract.
மதிப்பு	Dignified	No condescension. Big targets, plain words, real receipts. No marketing language; no decorative SVGs.

3 Tokens

3.1 Colour palette

Three theme variants share the same token names; only the values change. The default theme is **civic**; **paper** is a warmer alternate; **ink** is the dark variant. Set on `<html>` via `data-theme`.

Token	Civic value	Use
--paper	#FFFFFF	Card surface.
--paper-2	#F7F4EE	Page background — warm off-white.
--paper-3	#EFEAE0	Inset / chip / panel.
--rule	#DCD5C7	Hairline borders, dividers.
--ink	#14140F	Primary text. 17.8:1 against --paper-2 (AAA).
--ink-2	#3B3A33	Secondary text.
--ink-3	#6B6859	Metadata, captions.
--primary	#0E4D3F	Civic green. CTAs, OK status, primary actions. 11.2:1 (AAA).

Token	Civic value	Use
--accent	#D9A441	Saffron-gold. Verification, official seals, hero dot. 3.1:1 (AA-large).
--alert	#A6363D	Deep red. Alerts, destructive actions, rejected. 6.8:1 (AAA).
--info	#2B4A6B	Ink slate. Information, secondary status. 9.4:1 (AAA).

Each non-neutral colour has a -soft companion (light tint, used as badge background) and an -ink companion (dark variant, used as badge text). Example: --accent-soft: #F8EDD3, --accent-ink: #6B4F14.

The full token set (including --rule-strong, --ink-mute, --primary-soft, --primary-ink, --alert-soft, --alert-ink, --info-soft) is declared in frontend/src/app.css. Tailwind v4 @theme inline exposes them as utility tokens (bg-paper-2, text-ink-3, bg-accent-soft, etc.).

Rule: never hex-code colours in templates. Always use the token, either as a CSS variable (color: var(--ink)) or a Tailwind utility (text-ink).

3.2 Typography

Three font families, each with a clear role:

Family	Token	Used for
Noto Sans Tamil	--f-tamil	Body, headings, all user-facing copy. Primary face. Weights 300–800.
Inter	--f-latin	Latin numerals, English transliterations, EN sub-labels.
JetBrains Mono	--f-mono	Section tags, metadata captions, IDs, receipts, SMS bodies, technical labels. Conveys “auditable / transparent”.

Type scale (display → caption):

Role	Size	Weight	Use
Display	72px	700	Hero only. The “வெளி.” mark on the landing page.
Heading 1	36px	600	Page titles. Section titles in the design-system page.
Heading 2	24px	600	Card titles, sub-section headings.
Title	18px	600	In-card titles, list-row titles.
Body	16px	400	Default copy.
Caption	13px	500	Helper text, hints, errors.
Mono · meta	11px	500	Section tags, metadata, IDs, technical labels.

3.3 Spacing and radius

Spacing scale (for components and layout): xs 4 · sm 8 · md 14 · lg 20 · xl 32 · 2xl 48 · 3xl 72. Density is variable: data-density on <html> toggles compact / regular / comfy, which scales the row heights and paddings of list rows and similar components.

Radius scale: xs 4 · sm 6 · md 10 · lg 14 · xl 20 · pill 999. Cards use `--r-md`; pills use `--r-pill`; inputs use `--r-sm`; large CTAs use `--r-md`.

3.4 Elevation

Three shadow tiers, very restrained — paper-feeling, never floating. `--sh-1` is a hairline. `--sh-2` adds a 1px y-offset. `--sh-3` is the raised state for popovers. Avoid heavier shadows.

4 Components

Every primitive lives at `frontend/src/lib/components/ui/<Name>.svelte` and is re-exported from the barrel `frontend/src/lib/components/ui/index.ts`. Pages import the primitive, never the underlying CSS classes when a primitive applies.

Primitive	Purpose
Button	Variants <code>primary</code> / <code>secondary</code> / <code>ghost</code> / <code>alert</code> ; sizes <code>sm</code> / <code>md</code> / <code>lg</code> ; <code>block</code> . 48 px min height (≥ 44 WCAG) at <code>md</code> . Composes the <code>.btn</code> class family.
Card + CardHeader + CardContent + CardFooter	Raised paper surface with hairline border, optional header / content / footer. Composes <code>.card</code> .
Badge	Inline pill for short generic state. Variants <code>primary</code> / <code>accent</code> / <code>alert</code> / <code>info</code> / <code>mute</code> . Optional dot.
TierBadge	Three-tier verification visual. Always renders both the EN and TA tier labels — never one without the other. Visual hierarchy follows legal effect: <code>self_asserted</code> → bronze, <code>community_corroborated</code> → silver, <code>authority_attested</code> → gold.
StatusStrip	Thin bar reporting backend reachability and phase number. Civic green when reachable; alert-soft when unreachable.
LocaleSwitcher	Three-button segmented control (தமிழ் / English / [?] [?] [?]) calling Paraglide <code>setLocale</code> . Composes <code>.seg</code> .
Field	Label + input + (hint or error) wrapper. The <code><input></code> / <code><select></code> / <code><textarea></code> is supplied by the consumer with <code>class="field-input"</code> .
LRow	The workhorse list row. Optional leading icon, title, meta, optional EN sub-label, optional trail (snippet). Used for procedure lists, document evidence, buyer offers, user fields.
Icon	Line icon (lucide-svelte under the hood). 24 px viewBox, 1.6 px stroke, <code>currentColor</code> . Filled variant available via <code>filled</code> prop.
OfflinePill	Saffron pill that reads “இணையமில்லை · OFFLINE”. Always-visible signal when the service worker is serving cached content.
Progress	Slim progress bar in civic green on a paper-3 track. Used for evidence completeness, sync progress.
Segmented	Generic segmented-control primitive (the same <code>.seg</code> used by <code>LocaleSwitcher</code>).

4.1 TierBadge contract

The three tiers are normative. Their values, English labels, Tamil labels, and visual treatment are listed below. Never substitute, paraphrase, or invent a fourth.

Tier value	EN	TA	Visual
self_asserted	Self-asserted	சுய அறிவிப்பு	Bronze (tier-bronze) – visibly lowest-effort.
community_corroborated	Community-corroborated	சமூக சாட்சியம்	Silver (tier-silver) – neutral, no claim of legal effect.
authority_attested	Authority-attested	அதிகாரப்பூர்வ உறுதி	Gold (tier-gold) – saffron-soft tones, used only when the tier carries legal effect.

The dual-language label is the architectural contract from docs/plan/02_phase0_foundations.typ. A surface that needs to express a **derived** state (e.g. “expired attestation”) does so with a separate adjacent badge – never by changing the tier label.

5 Patterns

5.1 Hero

The landing page uses the giant Tamil wordmark with a saffron dot accent: `<h1 class="hero-mark">வெளி<span class="dot">.</span></h1>. clamp(72px, 12vw, 168px)` so it scales from phone to desktop. Bilingual metadata (`<dl class="hero-meta">`) sits beside or below the mark with `dt` in mono uppercase and `dd` in body.

5.2 Page shell

Every page wraps content in:

```
<div class="flex min-h-screen flex-col bg-background text-foreground">
  <main class="mx-auto w-full max-w-2xl px-6 py-12">
    <!-- page body -->
  </main>
</div>
```

Landing widens to `max-w-5xl`. User detail uses `max-w-2xl`. Avoid full-bleed on phones.

5.3 Section header

```
<div class="sec-tag">06 · கூறுகள் / Components</div>
<h2 class="text-[36px] font-semibold leading-tight tracking-tight">...</h2>
<p class="text-ink-2 max-w-[64ch]">...</p>
```

The mono section tag with the civic-green dot is the recurring section marker. Use bilingual `NN · தமிழ் /` English form when the section has a Tamil concept name.

5.4 List of properties

LRow rows inside a Card are the canonical way to display record fields (user details, document metadata, agricultural batch detail). Lead icon (40 px tile, paper-3 background) + title + meta + optional trail. Don’t use `<dl>` for record displays in v0.2 — LRow is the dignified pattern.

5.5 Trust / citation

Civic claims that come from a government source render as a Trust component (when added — currently in the design-system catalogue but not yet a primitive in this codebase). Until it lands, follow the markup directly: `.trust` wrapper + `.trust-mark` (32 px primary circle with shield icon) + title + body + `.trust-cite` (mono caption prefixed with ௩). Cite the source URL or the in-platform audit-trail reference, never an unverifiable claim.

5.6 Receipt

Confirmation numbers and submission receipts use the Receipt component (CSS class `.receipt`). Mono font, dashed top/bottom borders, the receipt-num as the centred final line. The user can screenshot it; SMS fallback delivers the same number.

5.7 SMS / USSD fallback

When bandwidth dies, surfaces fall back to SMS over the Veli-controlled shortcode (Ideamart). The SMS component shows the message as it will appear on a feature phone. Mono everywhere; preserves whitespace.

5.8 Form inputs

Every input is wrapped in a Field primitive: label above (never placeholder-only), `.field-input` for the control (full-width on mobile, 48 px tall, 1.5 px border), helper text below in `.field-help`, errors in `.field-error` (with the auto-rendered ! icon). Required fields show a red * in the label.

5.9 Empty + error states

Localised, never English-only. Use a Card with centred copy:

```
<Card class="text-center">
  <CardContent>{m.user_not_found()}</CardContent>
</Card>
```

Server errors use `border-alert/30 bg-alert-soft text-ink` plus a single line of guidance (“please try again later” — never a stack trace). Never expose raw backend error envelopes to the user; log them.

6 Accessibility

- **Touch targets:** 44 px minimum. `Button size="md"` is 48 px.
- **Focus rings:** never disable. The `--ring` token is `--primary`; a 2 px ring at 2 px offset is the default focus halo.
- **Contrast:** civic palette body sizes meet AA. Tier badges and the saffron offline pill have been verified against the relevant backgrounds; consult the colour table when adding new combinations.
- **aria-label:** every interactive control needs one, **and it must be localised through Paraglide**. Never hard-code English.
- **Language tag:** `<html lang>` is set by Paraglide via the `%paraglide.lang%` placeholder. Don’t override.

7 Internationalisation

- **Tamil first.** Author the Tamil string before any other language. Never machine-translate Tamil from English in user-facing copy.
- **Register.** Northern Province colloquial (Jaffna). Native review is in the loop and authoritative.
- **Catalogue.** All keys in `frontend/messages/{en,ta,si}.json`, one key per UI string. Pluralisation and parameter substitution use Paraglide’s parameter syntax — no string concatenation across keys.
- **Numerals.** Use European numerals (0–9) by default; Tamil numerals only when explicitly requested by content.

8 Mobile-first defaults

- Default breakpoints follow Tailwind: `sm` 640 px, `md` 768 px, `lg` 1024 px. The smartphone surface targets `< sm`; widen up from there.
- Never assume hover. Every interactive state must work on touch.
- Service worker caches the active session; pages render meaningful content with cached data plus a small banner when offline. Avoid spinner-only offline states.
- Capacitor Android wrap: status-bar colour is `bg-paper-2`; splash hidden on layout mount.

9 Code conventions

- **Svelte 5 runes only.** `$state`, `$derived`, `$effect`, `$props`.

- **Server-only** code lives in `src/lib/server/`. Never import in a `.svelte` component.
- **API calls** go through `+page.server.ts load()` (or form actions), using the typed client at `$lib/api`. Never direct browser fetch.
- **Primitive imports.** Reach for `$lib/components/ui/<Name>.svelte` before writing raw Tailwind for buttons, cards, badges, list rows, tier badges, status strips, fields, locale switchers, segmented controls, progress bars, offline pills, or icons.
- **cn() helper.** Compose classes with `cn(...)` from `$lib/utils`.
- **Props.** Type with TypeScript interfaces; default values via destructuring at the `$props()` site.
- **No inline <style>** unless scoped and unavoidable. Tailwind utilities first; CSS classes from `app.css` for composed components; CSS variables when a token is needed.

10 Boundaries

This design system covers the smartphone surface (PWA + Capacitor Android wrap). It does **not** cover:

- The IVR / USSD / SMS surfaces (governed by Tamil voice-prompt authoring rules, see strategic plan \$Phase 0 architecture).
- Internal admin or back-office tooling (out of scope for v1.x).
- Print or PDF artefacts (deed-ready bundles in Phase 2 use a separate Typst template).

When in doubt, defer to `CLAUDE.md` for project-wide rules and the strategic plan in `docs/plan/` for product policy.